Toy car loop-the-loop

Request: I want a toy car that does a loop

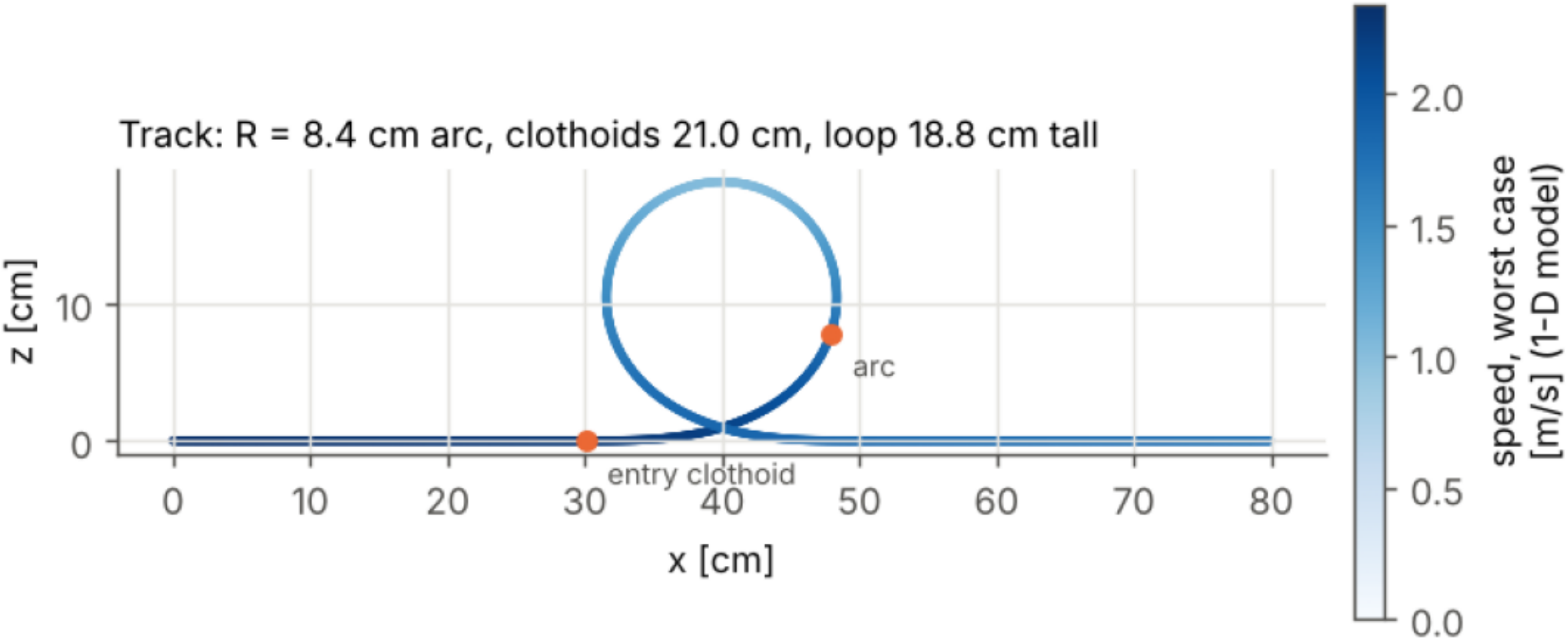
Verdict: will work   P = 1.000 (95 % CI 0.998-1.000)   trust: APPROVED

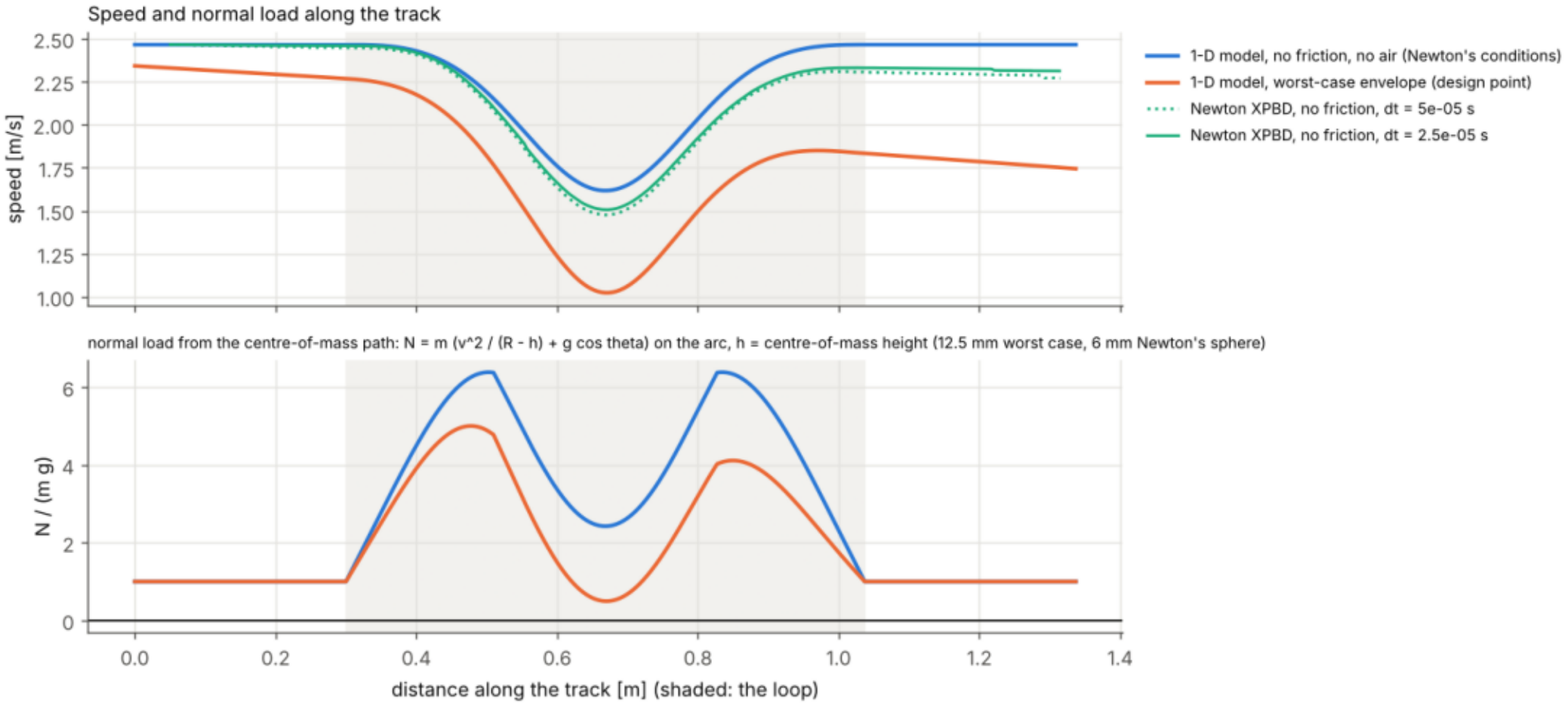
Design:
   loop radius R: 0.08392 m
   clothoid length / R: 2.498
   launch speed: 2.465 m/s
   car mass: 0.03101 kg

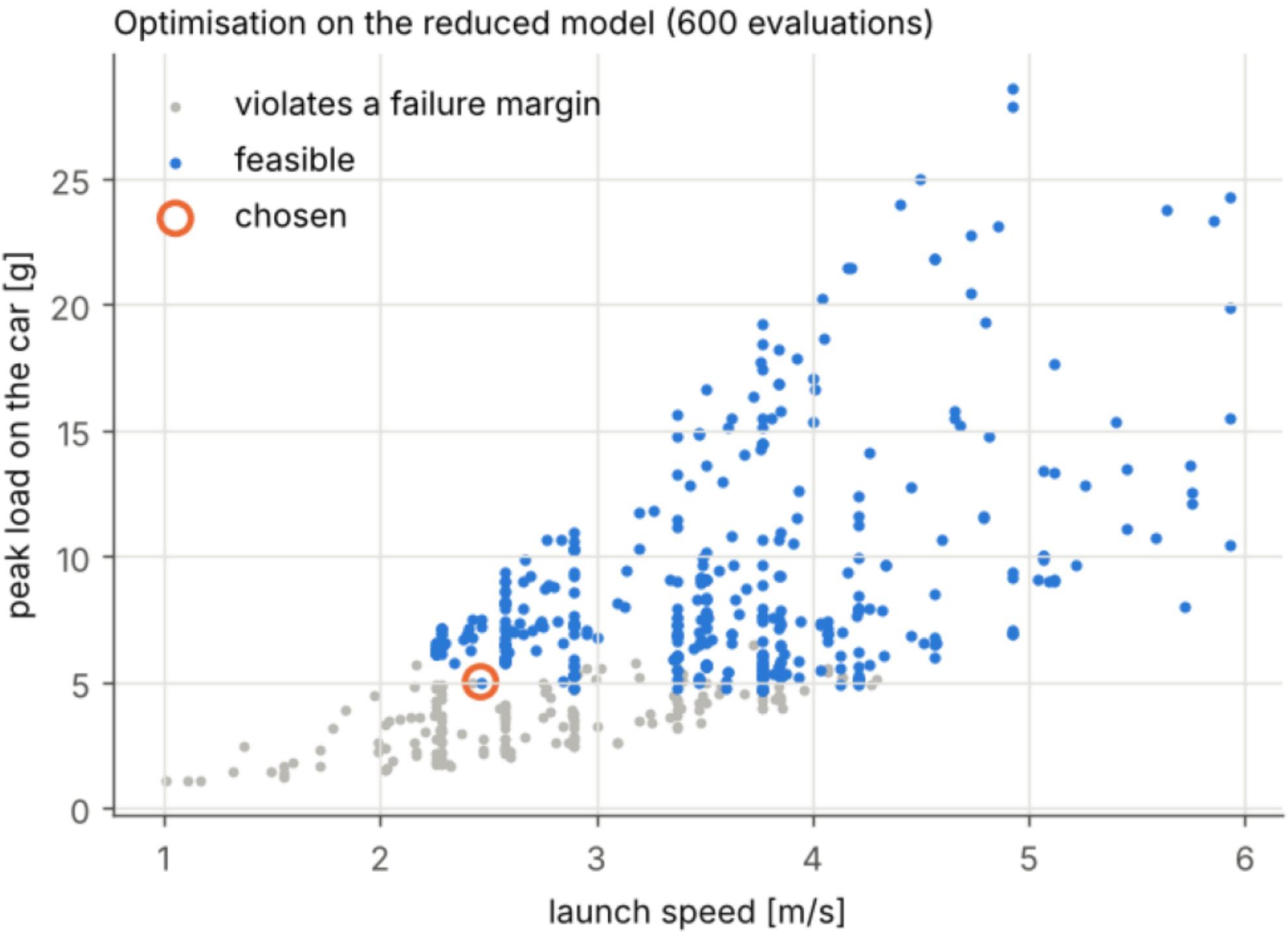
Numbers (fidelity):
   thickness: 0.0015 m [1-D along-track ODE]
   launch_speed: 2.465 m/s [1-D along-track ODE]
   footprint: 0.7992 m [1-D along-track ODE]
   loop fits the space available: 1   [1-D along-track ODE]
   track strip does not yield: 10.84   [1-D along-track ODE]
   track_mass_kg: 0.1467 kg [1-D along-track ODE]
   v_top: 1.023 m/s [1-D along-track ODE]
   car body clears the track in the loop: 1.007   [1-D along-track ODE]
   g_peak: 4.999 g [1-D along-track ODE]
   loop_radius: 0.08392 m [1-D along-track ODE]
   track_stress_mpa: 2.534 MPa [1-D along-track ODE]
   car stays on the track: 0.495   [1-D along-track ODE]
   loop_height: 0.2085 m [1-D along-track ODE]
   car stays on the track: 0.4927   [energy balance]
   track strip does not yield: 10.94   [energy balance]
   car body clears the track in the loop: 1.007   [energy balance]
   loop fits the space available: 1   [energy balance]
   g_peak: 4.955 g [energy balance]
   v_top: 1.023 m/s [energy balance]
   v_top: 1.507 m/s [Newton rigid-body contact]
   completes: 1   [Newton rigid-body contact]
   max_gap_m: 6.331e-05   [Newton rigid-body contact]
   v_top_1d_no_losses: 1.617   [Newton rigid-body contact]
   newton_energy_lost: 0.0572   [Newton rigid-body contact]

Verification:
   v_top: full 1.507, reduced 1.023, rel. err 0.3208

Limits:
   - The car is a point on its centre-of-mass path (1-D model) or a sphere (Newton); wheel bounce, steering and track joints are not modelled.
   - Newton's check has no friction and no air, and its contact dissipates energy: it confirms the 3-D track lets the car go round with margin, not the 1-D model's speeds to a tolerance.
   - Rolling resistance and drag are envelopes, not measurements: the design covers them, a car outside them (bent axle, dirty wheels) may not complete the loop.
   - The track strength check is a beam under a point load between clamps; the clamps, the stand and fatigue of the strip are not checked.
   - Toy scale only: nothing here applies to rides that carry people.


Track: R = 8.4 cm arc, clothoids 21.0 cm, loop 18.8 cm tall
z [cm]
10
0
arc
entry clothoid
0
10
20
30
40
50
60
70
80
x [cm]
2.0
1.5
1.0
0.5
0.0
speed, worst case
[m/s] (1-D model)


Speed and normal load along the track
speed [m/s]
2.50
2.25
2.00
1.75
1.50
1.25
1.00
1-D model, no friction, no air (Newton's conditions)
1-D model, worst-case envelope (design point)
Newton XPBD, no friction, dt = 5e-05 s
Newton XPBD, no friction, dt = 2.5e-05 s
normal load from the centre-of-mass path: N = m (v^2 / (R - h) + g cos theta) on the arc, h = centre-of-mass height (12.5 mm worst case, 6 mm Newton's sphere)
N / (m g)
6
4
2
0
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
distance along the track [m] (shaded: the loop)


Optimisation on the reduced model (600 evaluations)
violates a failure margin
feasible
chosen
peak load on the car [g]
launch speed [m/s]
0
5
10
15
20
25
1
2
3
4
5
6

